Negative Binomial and Zero-Inflated Negative Binomial Regression from Scratch: the nbzinbCirkovic Package

STA 651 Final Project, Fall 2026

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1 Introduction

Count outcomes, such as the number of physician visits or the number of published articles, are often modelled with Poisson regression. The Poisson model forces the variance to equal the mean. Real count data usually violate this in two ways: the variance is much larger than the mean (**overdispersion**), and there are more zeros than a Poisson distribution with the same mean would produce (**excess zeros**). Ignoring overdispersion makes standard errors too small and tests anti-conservative.

The negative binomial (NB) model adds a dispersion parameter θ that lets the variance exceed the mean. The zero-inflated negative binomial (ZINB) model goes further and treats the data as a mixture of a group that always produces zeros and a group that follows an NB distribution.

The goal of this project is an R package, **nbzinbCirkovic**, that fits both models *without* any general-purpose optimizer. NB regression is fitted by iteratively reweighted least squares (IRLS) alternated with Newton–Raphson for $\log\theta$, and ZINB regression by an EM algorithm whose M-steps are IRLS fits. We show that the package reproduces `MASS::glm.nb()` and `pscl::zeroinfl()`, check by simulation that `zinbreg()` recovers known parameters, and use it to analyze physician office visits in the NMES1988 data.

2 Methods

2.1 The negative binomial model

Let Y_i be a count with covariate vector x_i, $i = 1, \dots, n$. The NB2 model is

$$P(Y_i = y) = \frac{\Gamma(y+\theta)}{\Gamma(\theta)\, y!} \left(\frac{\theta}{\theta+\mu_i}\right)^{\theta} \left(\frac{\mu_i}{\theta+\mu_i}\right)^{y}, \qquad \log\mu_i = \eta_i = x_i^{\top}\beta + o_i,$$

with optional offset o_i. Then $E(Y_i) = \mu_i$ and $\mathrm{Var}(Y_i) = \mu_i + \mu_i^2/\theta$; as $\theta \to \infty$ the model reduces to the Poisson. With prior weights a_i the log-likelihood is

$$\ell(\beta, \theta) = \sum_i a_i \Big[\log\Gamma(y_i + \theta) - \log\Gamma(\theta) - \log y_i! + \theta \log\theta + y_i \log\mu_i - (\theta + y_i)\log(\theta + \mu_i) \Big].$$

IRLS for β (θ fixed). For fixed θ the NB is an exponential-family GLM with variance function $V(\mu) = \mu + \mu^2/\theta$. Fisher scoring is equivalent to repeated weighted least squares with

$$w_i = a_i \frac{(\partial\mu_i/\partial\eta_i)^2}{V(\mu_i)} = a_i \frac{\theta\mu_i}{\theta + \mu_i}, \qquad z_i = (\eta_i - o_i) + \frac{y_i - \mu_i}{\mu_i}, \qquad \beta^{(t+1)} = (X^\top W X)^{-1} X^\top W z.$$

The weighted least-squares step is solved with a QR decomposition of $W^{1/2}X$, and iterations stop when $\max_j |\beta_j^{(t+1)} - \beta_j^{(t)}| < 10^{-8}$.

Newton–Raphson for θ (β fixed). The score and second derivative are

$$U(\theta) = \sum_i a_i \Big[\psi(y_i + \theta) - \psi(\theta) + \log\theta + 1 - \log(\theta + \mu_i) - \frac{y_i + \theta}{\theta + \mu_i} \Big],$$

$$H(\theta) = \sum_i a_i \Big[\psi'(y_i + \theta) - \psi'(\theta) + \frac{1}{\theta} - \frac{2}{\theta + \mu_i} + \frac{y_i + \theta}{(\theta + \mu_i)^2} \Big].$$

We update $\phi = \log\theta$, which keeps $\theta > 0$: with $U_\phi = \theta U$ and $H_\phi = \theta^2 H + \theta U$, $\phi \leftarrow \phi - U_\phi / H_\phi$. Each step is halved until the log-likelihood does not decrease. In the code, U and H are evaluated in an algebraically equivalent form (for example $\log\theta - \log(\theta + \mu) = -\log(1 + \mu/\theta)$) that avoids cancellation when θ is large.

Outer loop and standard errors. Starting from a Poisson IRLS fit and the moment estimate $\theta_0 = n / \sum_i (y_i/\mu_i - 1)^2$, the two updates are alternated until the log-likelihood changes by less than 10^{-8}. At convergence $\widehat{\mathrm{Var}}(\hat\beta) = (X^\top \hat W X)^{-1}$, treating θ as known as `glm.nb()` does (this is justified because β and θ are orthogonal: $E[\partial^2\ell/\partial\beta\,\partial\theta] = 0$), and $\mathrm{SE}(\hat\theta) = \{-H(\hat\theta)\}^{-1/2}$.

2.2 The zero-inflated negative binomial model

With probability π_i observation i is a structural zero; otherwise it is $\mathrm{NB}(\mu_i, \theta)$:

$$P(Y_i = 0) = \pi_i + (1 - \pi_i) f_{NB}(0; \mu_i, \theta), \qquad P(Y_i = y) = (1 - \pi_i) f_{NB}(y; \mu_i, \theta),\ y \geq 1,$$

with $f_{NB}(0; \mu, \theta) = \{\theta/(\theta + \mu)\}^\theta$, $\log\mu_i = x_i^\top\beta + o_i$ and $\mathrm{logit}\,\pi_i = z_i^\top\gamma$. The marginal mean is $E(Y_i) = (1 - \pi_i)\mu_i$. The observed-data log-likelihood

$$\ell(\beta, \gamma, \theta) = \sum_{y_i = 0} \log\{\pi_i + (1 - \pi_i) f_{NB}(0; \mu_i, \theta)\} + \sum_{y_i > 0} \{\log(1 - \pi_i) + \log f_{NB}(y_i; \mu_i, \theta)\}$$

contains the logarithm of a sum, so it is not an exponential-family likelihood and cannot be maximized by IRLS directly.

EM algorithm. Let $D_i = 1$ if observation i is a structural zero. If the D_i were known, the complete-data log-likelihood would split into a logistic-regression part in γ and a weighted NB part in (β, θ). EM replaces D_i by its conditional expectation.

- *E-step:* $\tau_i = P(D_i = 1 \mid y_i) = \pi_i / \{\pi_i + (1-\pi_i) f_{NB}(0; \mu_i, \theta)\}$ if $y_i = 0$, and $\tau_i = 0$ if $y_i > 0$.
- *M-step, zero part:* maximize $\sum_i \tau_i \log \pi_i + (1-\tau_i) \log(1-\pi_i)$, a logistic regression with fractional response, by IRLS with $w_i = \pi_i(1-\pi_i)$ and $z_i = z_i^\top \gamma + (\tau_i - \pi_i)/\{\pi_i(1-\pi_i)\}$.
- *M-step, count part:* maximize $\sum_i (1-\tau_i) \log f_{NB}(y_i; \mu_i, \theta)$, which is exactly the NB fit above with prior weights $a_i = 1 - \tau_i$.

Both M-steps are iterated to convergence, so each EM iteration cannot decrease ℓ; the package records ℓ at every iteration and warns if it ever decreases. Starting values are the NB fit for (β, θ) and a logistic IRLS fit of $I(y_i = 0)$ on z_i for γ. EM stops when ℓ changes by less than 10^{-8}.

Standard errors. EM does not provide an information matrix, so we invert the observed information $-\nabla^2 \ell$ with respect to $(\beta, \gamma, \log\theta)$ at the estimates. The Hessian is computed **analytically**. Because ℓ_i depends on β only through $\eta_i = x_i^\top \beta + o_i$ and on γ only through $\zeta_i = z_i^\top \gamma$, every block is a weighted cross-product, e.g. $\partial^2 \ell / \partial\beta\, \partial\gamma^\top = X^\top \operatorname{diag}(\ell_{\eta\zeta,i}) Z$, and only six scalar second derivatives per observation are needed. For $y_i > 0$ the zero and count parts separate, so $\ell_{\eta\zeta} = \ell_{\zeta\phi} = 0$, $\ell_{\zeta\zeta} = -\pi_i(1-\pi_i)$ and $\ell_{\eta\eta} = -\theta\mu_i(\theta + y_i)/(\theta + \mu_i)^2$. For $y_i = 0$, $\ell_i = \log L_i$ with $L_i = \pi_i + (1-\pi_i) f_0$ and $\ell_{ab} = L_{ab}/L - L_a L_b / L^2$. The analytic Hessian agrees with a numerical Hessian (`numDeriv`) to a relative error below 10^{-6} (tested in the package). $\mathrm{SE}(\hat\theta)$ follows from the delta method, $\hat\theta\, \mathrm{SE}(\log\hat\theta)$.

3 The R package

The package exports two fitting functions and their control functions:

- `nbreg(formula, data, offset, control = nbreg_control())`
- `zinbreg(formula, data, offset, control = zinbreg_control())`, with a two-part formula `y ~ x1 + x2 | z1 + z2` (count covariates before `|`, zero-model covariates after; without `|` both parts use the same covariates).

Internally, `R/irls.R` holds the IRLS and Newton–Raphson building blocks, `nb_fit()` the NB outer loop (reused, with weights $1 - \tau_i$, as the ZINB count M-step), `zinb_fit()` the EM loop, and `R/hessian.R` the analytic derivatives. Both functions return S3 objects (`"nbreg"`, `"zinbreg"`) with methods `print`, `summary`, `coef`, `vcov`, `logLik` (with a `df` attribute, so `AIC()` and `BIC()` work), `fitted`, `residuals` (response and Pearson), `nobs` and `predict` (with `newdata`; for `zinbreg`, `type = "response"`, `"count"` or `"zero"`). The control functions set tolerances, iteration limits and a `trace` option; the fitting functions warn when the algorithm does not converge or when $\hat\theta$ is very large (the Poisson limit).

The package has roxygen2 documentation with runnable examples, four testthat files that compare it with `glm.nb()` and `zeroinfl()` and check offsets, single-covariate models, error messages and the monotone EM log-likelihood, and a GitHub Actions workflow running `R CMD check` on Linux, macOS and Windows. Installation and a short example:

```
# install.packages("devtools")
devtools::install_github("cirkovicdavid/nbzinbCirkovic")
library(nbzinbCirkovic)
fit <- zinbreg(art ~ fem + mar + kid5 + phd + ment | ment, data = bioChemists)
```

```r
summary(fit)
predict(fit, newdata = bioChemists[1:3, ], type = "zero")
```

4 Benchmarking

We fitted both models to `pscl::bioChemists` (art ~ fem + mar + kid5 + phd + ment, all covariates in both ZINB parts) and `AER::NMES1988` (the specification of Section 6) with our package and with the reference functions, run with tight tolerances (`glm.control(epsilon = 1e-10)`, `zeroinfl.control(reltol = 1e-12)`). Tables 1–4 show the estimates side by side; Table 5 summarizes the maximum absolute differences against the required tolerances.

Table 1: bioChemists, NB: nbreg() vs glm.nb().

Term	Ours	Reference	SE ours	SE ref.	|diff| est.	|diff| SE
(Intercept)	0.25614	0.25614	0.13735	0.13735	8.1e-10	5.2e-10
femWomen	-0.21642	-0.21642	0.07264	0.07264	1.6e-10	2.7e-10
marMarried	0.15049	0.15049	0.08210	0.08210	6.5e-11	3.1e-10
kid5	-0.17642	-0.17642	0.05281	0.05281	1.3e-10	1.9e-10
phd	0.01527	0.01527	0.03587	0.03587	5.5e-11	1.4e-10
ment	0.02908	0.02908	0.00321	0.00321	5.7e-11	1.6e-11
theta	2.26439	2.26439	0.27118	0.27118	4.0e-08	1.1e-06

Table 2: bioChemists, ZINB: zinbreg() vs zeroinfl().

Term	Ours	Reference	SE ours	SE ref.	|diff| est.	|diff| SE
count_(Intercept)	0.41675	0.41675	0.14360	0.14360	2.3e-06	3.3e-07
count_femWomen	-0.19550	-0.19551	0.07559	0.07559	2.3e-06	1.2e-07
count_marMarried	0.09758	0.09758	0.08445	0.08445	1.3e-06	1.9e-08
count_kid5	-0.15173	-0.15173	0.05421	0.05421	1.7e-06	1.3e-08
count_phd	-0.00070	-0.00070	0.03627	0.03627	6.5e-07	7.3e-08
count_ment	0.02479	0.02479	0.00349	0.00349	3.1e-07	2.6e-07
zero_(Intercept)	-0.19184	-0.19169	1.32272	1.32282	1.5e-04	9.4e-05
zero_femWomen	0.63595	0.63593	0.84885	0.84892	1.2e-05	6.6e-05
zero_marMarried	-1.49932	-1.49947	0.93859	0.93867	1.5e-04	8.2e-05
zero_kid5	0.62841	0.62843	0.44276	0.44278	2.1e-05	2.4e-05
zero_phd	-0.03768	-0.03771	0.30799	0.30801	3.3e-05	2.1e-05
zero_ment	-0.88222	-0.88229	0.31619	0.31623	7.6e-05	3.6e-05
Log(theta)	0.97637	0.97636	0.13547	0.13547	1.3e-05	4.3e-07

Table 3: NMES1988, NB: nbreg() vs glm.nb().

Term	Ours	Reference	SE ours	SE ref.	|diff| est.	|diff| SE
(Intercept)	0.92926	0.92926	0.05459	0.05459	4.7e-10	5.9e-12
hospital	0.21777	0.21777	0.02018	0.02018	3.5e-10	2.3e-12
healthpoor	0.30501	0.30501	0.04851	0.04851	3.1e-10	5.1e-12
healthexcellent	-0.34181	-0.34181	0.06092	0.06092	8.9e-11	6.3e-12
chronic	0.17492	0.17492	0.01209	0.01209	1.4e-10	1.3e-12
gendermale	-0.12649	-0.12649	0.03122	0.03122	5.7e-11	3.3e-12
school	0.02682	0.02682	0.00439	0.00439	6.1e-12	4.7e-13
insuranceyes	0.22440	0.22440	0.03946	0.03946	8.3e-11	4.3e-12
theta	1.20660	1.20660	0.03356	0.03355	3.6e-10	4.2e-07

Table 4: NMES1988, ZINB: zinbreg() vs zeroinfl().

Term	Ours	Reference	SE ours	SE ref.	|diff| est.	|diff| SE
count_(Intercept)	1.19372	1.19371	0.05666	0.05666	3.2e-06	6.5e-08
count_hospital	0.20148	0.20148	0.02036	0.02036	2.7e-08	1.2e-08
count_healthpoor	0.28513	0.28513	0.04509	0.04509	6.8e-08	4.2e-08
count_healthexcellent	-0.31934	-0.31934	0.06040	0.06040	6.5e-09	4.0e-08
count_chronic	0.12900	0.12900	0.01193	0.01193	4.5e-07	1.2e-08
count_gendermale	-0.08028	-0.08028	0.03102	0.03102	5.9e-07	4.1e-08
count_school	0.02142	0.02142	0.00436	0.00436	3.7e-09	1.6e-08
count_insuranceyes	0.12586	0.12586	0.04159	0.04159	1.6e-06	5.0e-08
zero_(Intercept)	-0.04689	-0.04688	0.26856	0.26856	9.0e-06	4.6e-07
zero_hospital	-0.80046	-0.80048	0.42080	0.42082	2.6e-05	1.7e-05
zero_chronic	-1.24791	-1.24794	0.17831	0.17832	2.6e-05	6.0e-06
zero_gendermale	0.64768	0.64768	0.20011	0.20011	2.5e-06	3.1e-06
zero_school	-0.08377	-0.08377	0.02625	0.02625	1.4e-06	4.3e-07
zero_insuranceyes	-1.17561	-1.17562	0.22012	0.22012	4.6e-06	4.5e-06
Log(theta)	0.39415	0.39414	0.03504	0.03504	4.4e-06	4.0e-08

Table 5: Maximum absolute differences from the reference functions (Theta: theta for NB, log theta for ZINB) and median computing times in seconds. Required: NB $\leq 10^{-5}$ (coefficients, SEs) and $\leq 10^{-4}$ (θ, log-likelihood); ZINB $\leq 10^{-3}$ for all quantities.

Data set	Model	Coefficients	Std. errors	Theta	Log-lik.	Ours (s)	Ref. (s)
bioChemists	NB	8.1e-10	5.2e-10	4.0e-08	1.4e-12	0.008	0.022
bioChemists	ZINB	1.5e-04	9.4e-05	1.3e-05	4.2e-08	0.388	0.049
NMES1988	NB	4.7e-10	6.3e-12	3.6e-10	4.2e-11	0.035	0.078
NMES1988	ZINB	2.6e-05	1.7e-05	4.4e-06	1.6e-08	0.820	0.229

All differences are well inside the tolerances of the assignment. The NB fits agree with `glm.nb()` to about 10^{-9}, and `nbreg()` (typically 3 outer iterations, because β and θ are orthogonal) is as fast

as or faster than `glm.nb()`. The ZINB fits agree with `zeroinfl()` to about 10^{-4}. EM needs 111 iterations on bioChemists and 53 on NMES1988 because it converges only linearly, so `zinbreg()` is several times slower than `zeroinfl()`'s BFGS, but still takes under one second. Our ZINB log-likelihoods are 4.2e-08 and 1.6e-08 from the reference values.

5 Simulation study

We simulated $R = 500$ data sets of size $n = 500$ from a ZINB model with $\log \mu_i = \beta_0 + \beta_1 x_i$, $\operatorname{logit} \pi_i = \gamma_0 + \gamma_1 z_i$, $x_i, z_i \sim N(0,1)$ independent, and $(\beta_0, \beta_1, \gamma_0, \gamma_1, \theta) = (1, 0.5, -1, 0.8, 2)$ (seed 20261031). On average 43% of the responses were zero. `zinbreg()` converged in all 500 data sets (34 seconds in total).

Table 6: Simulation results (R = 500, n = 500): bias, mean estimated SE vs. Monte Carlo SD, and empirical coverage of 95% Wald intervals.

Parameter	True	Mean est.	Bias	Mean SE	MC SD	Coverage 95%
β_0	1.000	0.995	-0.0053	0.0654	0.0644	0.944
β_1	0.500	0.496	-0.0037	0.0548	0.0599	0.926
γ_0	-1.000	-1.019	-0.0189	0.2218	0.2196	0.954
γ_1	0.800	0.817	0.0167	0.1818	0.1840	0.950
$\log \theta$	0.693	0.734	0.0407	0.2152	0.2224	0.940

All estimates are close to the truth: the biases of the regression coefficients are below 0.02 in absolute value, and $\log \hat{\theta}$ has a small positive bias (0.041), as expected for a dispersion parameter in moderate samples. The mean analytic standard errors are close to the Monte Carlo standard deviations, so the Hessian-based SEs are reliable. Coverage is close to the nominal 95% (the Monte Carlo standard error of a coverage estimate is about 0.010); the lowest value, 0.926 for β_1, reflects a mean SE slightly smaller than the Monte Carlo SD.

6 Data analysis: NMES1988

The NMES1988 data record the number of physician office visits (`visits`) for 4,406 people aged 66 and over in the 1987–88 US National Medical Expenditure Survey (Deb and Trivedi, 1997). Count-model covariates are the number of hospital stays (`hospital`), self-perceived health (`health`: poor, average [reference], excellent), number of chronic conditions (`chronic`), `gender`, years of education (`school`) and private insurance (`insurance`). The zero model uses `hospital`, `chronic`, `gender`, `school` and `insurance`, as specified in the assignment.

6.1 Descriptive summary

The mean number of visits is 5.77, while the variance is 45.7, about 8 times the mean: strong overdispersion. 15.5% of the respondents (683 people) made no visit, whereas a Poisson distribution with the same mean predicts only 0.31%. Figure 1 compares the observed distribution with the frequencies expected under the three fitted models.

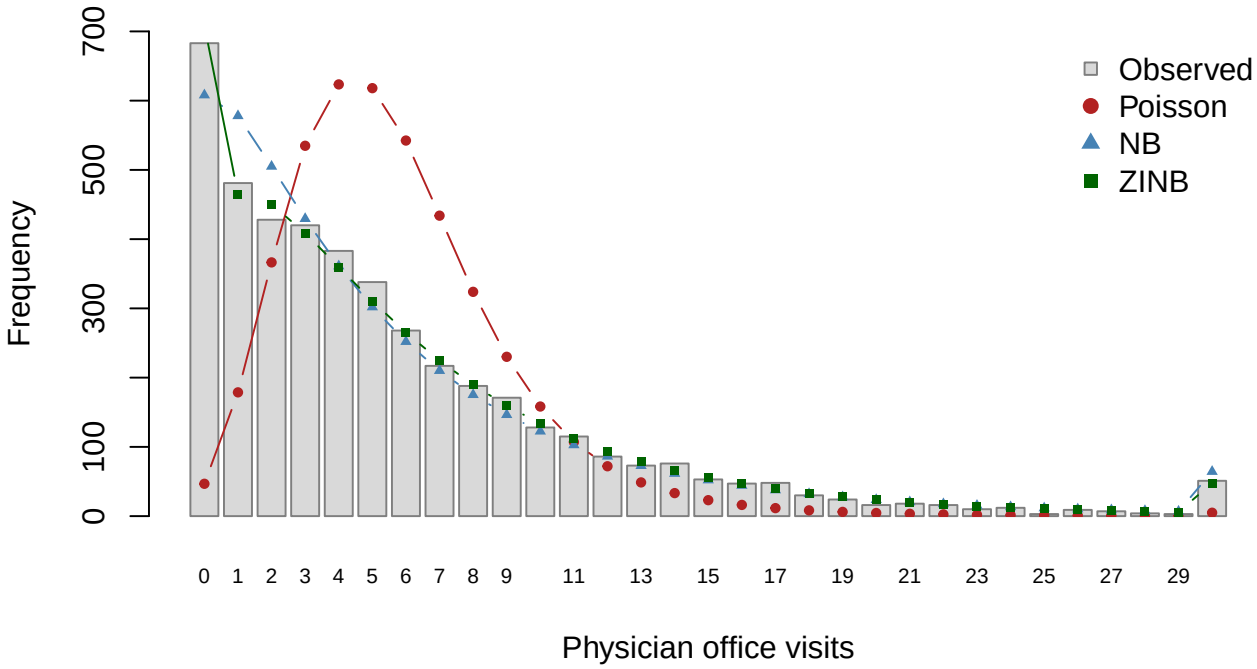


Figure 1: Observed frequencies of physician office visits (bars, 30 = 30 or more) and frequencies expected under the fitted Poisson, NB and ZINB models.

6.2 Model comparison

Table 7: Model comparison for NMES1988 (observed zeros = 683).

Model	df	Log-lik.	AIC	BIC	Expected zeros
Poisson (glm)	8	-17,971.6	35,959.2	36,010.4	46.7
NB (nbreg)	9	-12,170.6	24,359.1	24,416.6	608.0
ZINB (zinbreg)	15	-12,090.7	24,211.4	24,307.3	708.9

The Poisson model fits very poorly: its log-likelihood is about 5,801 lower than the NB model's, and it predicts only 47 zeros against 683 observed. Allowing overdispersion (NB, $\hat{\theta} = 1.21$) removes most of the lack of fit, but the NB model still under-predicts the zeros (608). The ZINB model improves the log-likelihood by 79.8 with 6 extra parameters, predicts 709 zeros, and has the lowest AIC **and** BIC.

6.3 Interpretation

Table 8: Count part: incidence rate ratios (IRR) with 95% Wald confidence intervals.

Covariate	NB IRR (95% CI)	ZINB count IRR (95% CI)	ZINB count p-value
hospital	1.24 (1.20, 1.29)	1.22 (1.18, 1.27)	<1e-04
healthpoor	1.36 (1.23, 1.49)	1.33 (1.22, 1.45)	<1e-04

Covariate	NB IRR (95% CI)	ZINB count IRR (95% CI)	ZINB count p-value
healthexcellent	0.71 (0.63, 0.80)	0.73 (0.65, 0.82)	<1e-04
chronic	1.19 (1.16, 1.22)	1.14 (1.11, 1.16)	<1e-04
gendermale	0.88 (0.83, 0.94)	0.92 (0.87, 0.98)	0.0097
school	1.03 (1.02, 1.04)	1.02 (1.01, 1.03)	<1e-04
insuranceyes	1.25 (1.16, 1.35)	1.13 (1.05, 1.23)	0.0025

Table 9: ZINB zero part: odds ratios (OR) of being a structural (never-visit) zero, with 95% Wald confidence intervals.

Covariate	OR (95% CI)	p-value
hospital	0.45 (0.20, 1.02)	0.0571
chronic	0.29 (0.20, 0.41)	<1e-04
gendermale	1.91 (1.29, 2.83)	0.0012
school	0.92 (0.87, 0.97)	0.0014
insuranceyes	0.31 (0.20, 0.48)	<1e-04

Dispersion. The ZINB model estimates $\hat{\theta} = 1.48$ (SE 0.052), so even among people who do visit a physician the counts are clearly overdispersed relative to the Poisson.

> **!** Important
>
> **TO DO (David — write in your own words, then delete this box):** interpret the main effects in plain language using Tables 8 and 9, as in Appendix A.4 of the project description. Cover (i) the count part as incidence rate ratios, e.g. what an IRR of 1.22 for `hospital` means for the expected number of visits, the effects of poor/excellent health versus average, chronic conditions, gender, education and insurance; and (ii) the zero part as odds ratios of being a structural "never-visit" zero: which covariates matter (chronic conditions, insurance, gender, education) and what their direction means for who never sees a physician.

6.4 Preferred model

> **!** Important
>
> **TO DO (David — write in your own words, then delete this box):** say which model you prefer and why, using Table 7 and Figure 1 (log-likelihood, AIC, BIC, observed vs. expected zeros), and contrast with the bioChemists example in Appendix A, where BIC preferred the NB model.

7 Discussion

The package shows that both models can be fitted with nothing more than weighted least squares, one-dimensional Newton–Raphson, and EM, and that the results match the established implementations

to within numerical tolerance. Several limitations remain. EM converges linearly, so `zinbreg()` needs tens to hundreds of iterations and is slower than a quasi-Newton maximizer; acceleration (e.g. SQUAREM) or switching to Newton steps on the full parameter vector near the optimum would speed it up. The NB standard errors for β treat θ as known, which is asymptotically valid but ignores estimation uncertainty in θ in small samples. The zero-part parameters can be weakly identified when few zeros are structural, leading to large standard errors (as in bioChemists). Extensions include prior weights and `subset`/`na.action` arguments for users, a zero-inflated Poisson option, hurdle models, robust (sandwich) standard errors, and offsets in the zero-inflation model.

8 AI-use statement

> **!** Important
>
> **TO DO (David — write this yourself, then delete this box).** Required by Section 6.1 item 8: how you used Claude Code (what you asked it to do), what you verified yourself, and one example where you had to correct or reject its output. A candidate example from this project: the first version of the NB log-likelihood was typed directly from the formula in Section 2.1; on Poisson-like data it suffered floating-point cancellation for large θ, so `nbreg()` failed to converge until the formula was rewritten with `dnbinom(log = TRUE)` and cancellation-free forms of U and H. Another: the first `devtools::check()` failed because the S3 methods did not import the `stats` generics.

9 References

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